One ebit is necessary and sufficient for the elegant joint measurement

Abstract

The classical-output elegant joint measurement has exact supplied-entanglement cost one ebit. The same cost holds when the measurement is approached by a sequence of physical LOCC protocols using the same finite pure resource. The protocols may have countably many turns, standard Borel outcomes, separable local workspaces, local mixed ancillary preparations, and a report defined on the entire transcript. The converse combines a separability witness, a determinant identity, and an LOCC concurrence bound. A finite-input approximation argument and compact augmented moments extend it to these protocols and their operational limits.

For one-way LOCC, the resource theorem of Bandyopadhyay, Halder, and Nathanson [6] already gives the one-ebit requirement for the elegant joint measurement. The converse below removes that restriction on communication and applies to the protocol class specified in this paper. Our cost is the entropy of the supplied finite pure resource. It differs from the cost studied by Bandyopadhyay, Brassard, Kimmel, and Wootters [7]: their definition averages the entanglement consumed and credits residual distillable entanglement, with outcome probabilities evaluated on the maximally mixed input. The present one-ebit equality concerns supplied entanglement, not that averaged net-consumption quantity.

1 Measurement, resources, and the protocol class

Let $A = B = \mathbb{C}^2$, let $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ be the Pauli matrices, and set

$$|s\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \qquad \mathbf{n}_j \in \frac{1}{\sqrt{3}}\{(1,1,1),(1,-1,-1),(-1,1,-1),(-1,-1,1)\}.$$

Choose one ordering of these four vectors. Define

$$|e_j\rangle = \left[\left(\frac{I}{2} + \frac{\sqrt{3}}{2}\mathbf{n}_j \cdot \boldsymbol{\sigma}\right) \otimes I\right] |s\rangle, \qquad P_j = |e_j\rangle\langle e_j|. \tag{1}$$

These projectors describe the elegant joint measurement [1]. Its shared classical-output channel is

$$\mathcal{M}_{\mathrm{EJM}}(\rho) = \sum_{j=1}^{4} \mathrm{Tr}(P_j\rho)|j\rangle\langle j|_{C_A} \otimes |j\rangle\langle j|_{C_B}. \tag{2}$$

The output registers are four dimensional. The required output in this paper is the channel (2).

A charged resource is any normalized pure state $|\chi\rangle \in A' \otimes B'$ with A', B' finite dimensional. Its cost is the entropy in bits of its entire Alice reduction,

$$E(\chi) = S(\chi_{A'}) = -\,\mathrm{Tr}(\chi_{A'} \log_2 \chi_{A'}), \qquad 0 \log_2 0 = 0.$$

There is no constraint on its finite Schmidt rank. All initial shared entanglement is included in this state. Independent local normal mixed ancillary states and local isometries are free.

We specify the physical LOCC class as follows. There are countably many chronological local turns. At each finite public history, the acting party is a measurable function of the history and applies a normal completely positive instrument whose total operation is trace preserving. The

outcome spaces are standard Borel and may depend on the history. The heterogeneous outcome records are measurably encoded at each turn in a standard Borel space, with a measurable admissible-history graph. The local working spaces are complete separable Hilbert spaces, allowed to change with the history and outcome. Their measurable Hilbert fields have countable dense fundamental sections. Instruments and coordinate wiring are measurable kernels: after measurable isometric embeddings into a fixed separable space, setwise instrument operations on trace-class inputs depend measurably on the history. These conditions include finite, countable, and standard continuous records and impose no bound on local workspace dimension. Normality means that the instruments act on trace-class states and have normal Heisenberg adjoints. This is the usual instrument setting [3, 2].

Each protocol has one input-independent report J taking values in $\{1, 2, 3, 4\}$ on the full common transcript. Stopping can be encoded by absorbing identity turns. A report on an infinite transcript is also allowed. The report may be measurable in the completion of the Born law of the prepared reference

$$\rho_* = V_\chi(I_{AB}/4)V_\chi^\dagger, \qquad V_\chi v = v \otimes |\chi\rangle, \tag{3}$$

with the tensor factors regrouped as $AA' : BB'$. It is reported to both parties. The physical outcomes, instruments, and wiring determine the Born law; the law is not additional protocol data.

Let $\mathcal{M}_{P,\chi}$ be the resulting shared report channel on AB. Define $\mathcal{C}_{\mathrm{ex}}$ to be the set of $E(\chi)$ for which one such protocol satisfies $\mathcal{M}_{P,\chi}(\rho) = \mathcal{M}_{\mathrm{EJM}}(\rho)$ for every density operator ρ on AB. Define $\mathcal{C}_{\mathrm{op}}$ by requiring instead protocols P_k with

$$\mathcal{M}_{P_k,\chi}(\rho) \longrightarrow \mathcal{M}_{\mathrm{EJM}}(\rho) \quad \text{for every density operator } \rho. \tag{4}$$

Here the entire charged resource χ stays fixed. The instruments, outcomes, workspaces, local mixed preparations, and completed reports may vary with k. All limits in (4) concern the fixed finite input and classical output spaces.

Theorem 1. *For this physical LOCC class, every exact or operationally implementing finite pure resource has $E(\chi) \geq 1$. A Bell pair admits a finite-turn exact implementation. Consequently*

$$\inf \mathcal{C}_{\mathrm{ex}} = \inf \mathcal{C}_{\mathrm{op}} = 1,$$

and both infima are attained.

2 Finite effects from physical protocols

The converse will use actual finite complete local Kraus trees. A node in such a tree has finitely many local Kraus operators with Gram sum I; each terminal occurrence has a report. Dimensions are fixed at the original finite local input dimensions.

Lemma 2. *Let the local input spaces be $\mathbb{C}^a : \mathbb{C}^b$, with $a, b > 0$. Every physical protocol with a Borel finite-valued report has report effects F_j satisfying $F_j \geq 0$ and $\sum_j F_j = I$. There are finite complete reported local Kraus trees with effects $G_{N,j}$ such that*

$$\max_j \|G_{N,j} - F_j\|_{\mathrm{op}} \longrightarrow 0.$$

Local normal mixed preparations and separable, history-dependent working spaces are included.

Proof. We give the construction, including the reduction to finite local inputs. Consider a history-dependent normal CP instrument Φ_h with finite domain $M_a(\mathbb{C})$ and output in the trace class of a separable Hilbert space G. For a measurable outcome set B, its positive Choi measure is

$$C_h(B) = \sum_{u,v=1}^{a} \Phi_h(B)(|u\rangle\langle v|) \otimes |u\rangle\langle v|, \qquad \mu_h(B) = \operatorname{Tr} C_h(B)/a.$$

The latter is a probability kernel. In a fixed countable basis of $G \otimes \mathbb{C}^a$, each scalar matrix-entry measure is absolutely continuous with respect to μ_h. Jointly Borel densities can be obtained by taking ratios on the atoms of increasing finite partitions generating the outcome sigma algebra and then taking their martingale limits. There are only countably many matrix entries. Outside a conditional null set, their limits form a positive matrix with diagonal sum a, hence a positive trace-class operator $D(h,x)$ with trace a. Set $D = 0$ on the exceptional set. Finite compressions show that D is trace-class-valued Borel, and uniform polynomial approximation to the square root on $[0,a]$ makes $D^{1/2}$ operator-norm Borel.

Unvectorize the columns $D^{1/2}e_\ell$ into maps $K_\ell(h,x) : \mathbb{C}^a \to G$. Equality of the Choi measures gives, for every complex matrix Z,

$$\Phi_h(B)(Z) = \int_B \sum_\ell K_\ell(h,x) Z K_\ell(h,x)^\dagger \, \mu_h(dx), \qquad \int \sum_\ell K_\ell^\dagger K_\ell \, d\mu_h = I. \tag{5}$$

The integrals converge in trace norm because $\sum_\ell \|K_\ell\|_{\mathrm{HS}}^2 = \operatorname{Tr} D \le a$. Choose $p_\ell > 0$ with $\sum_\ell p_\ell = 1$, replace the record by (x,ℓ), and use the density $K = K_\ell/\sqrt{p_\ell}$ against the probability kernel $\lambda_h(dx,\ell) = \mu_h(dx)p_\ell$. Then $\int K^\dagger K \, d\lambda_h = I$ and $\int \|K\|_{\mathrm{HS}}^2 \, d\lambda_h = a$. Revealing ℓ and allowing the continuation to ignore it recovers the original operation.

For varying Hilbert fibers, measurable Gram–Schmidt applied to their fundamental sections gives measurable embeddings into ℓ^2. The recorded output projection supports the Choi measure. Positivity therefore supports its density and every square-root column in the actual output fiber, almost everywhere. The preceding construction thus preserves the correct output space. Null branches admit an arbitrary complete continuation.

For every rectangular K, put $A = (K^\dagger K)^{1/2}$ and let V be its polar partial isometry. With $Q = I - V^\dagger V$ and a unit vector ϕ in the output space, the map

$$R_K(Z) = VZV^\dagger + \operatorname{Tr}(QZ)|\phi\rangle\langle\phi| \tag{6}$$

is normal CPTP and satisfies $R_K(AZA) = KZK^\dagger$. Its Kraus operators are V and $|\phi\rangle\langle i|Q$, whose Gram sum is I. Both V and its support projection are Borel functions of K: use the limits $K(K^\dagger K + t^{-1}I)^{-1/2}$ and $K^\dagger K(K^\dagger K + t^{-1}I)^{-1}$. This handles rank changes and $K = 0$. Zero-dimensional fibers carry only zero branches and can be replaced there by an absorbing one-dimensional fiber.

Carry two pending local CPTP maps from the fixed original finite input spaces into the actual working spaces. Initially these are the actual local preparation maps, including mixed preparations. When Alice acts, compose her instrument with her pending map and apply (5). In the compressed process she uses the square density A on $\mathbb{C}^a$ and replaces her pending map by R_K. Bob's map stays pending; the roles are reversed when he acts. The identity (6), including its reference extensions, proves branch equality after the pending maps. Local pending maps commute with remote operations. Induction proves equality of every finite-cylinder record probability. Consistency and the unique countable standard-Borel probability extension give the same full transcript law after the additional

Kraus labels are forgotten. Thus a measurable process with square local Kraus densities on the fixed finite spaces has exactly the original report probabilities.

Fix a finite horizon and a Borel report of its history. Backward recursion gives remaining report effects $E_j(h)$ with $0 \le E_j(h) \le I$ and $\sum_j E_j(h) = I$. At an Alice node, let $L_h(z)$ be the square density and $S_h(z) = L_h(z) \otimes I_b$. The recursion reads

$$E_j(h) = \int S_h(z)^\dagger E_j(h,z) S_h(z)\, \lambda_h(dz).$$

Sample the entire integrable finite-dimensional Hermitian tuple

$$(L_h(z)^\dagger L_h(z),\ S_h(z)^\dagger E_1(h,z) S_h(z), \dots, S_h(z)^\dagger E_q(h,z) S_h(z)).$$

An integrable finite-dimensional barycenter is in the closed convex hull of the values on every full-measure set: otherwise a separating linear functional contradicts its integral. Thus finitely many actual valid points z_i and positive weights w_i summing to one simultaneously approximate the tuple. For any $0 < \delta < 1$ they may be chosen with

$$\|Q - I\|_{\mathrm{op}} < \delta, \quad \|C_j - E_j(h)\|_{\mathrm{op}} < \delta, \quad Q = \sum_i w_i L_i^\dagger L_i, \quad C_j = \sum_i w_i S_i^\dagger E_j(h, z_i) S_i.$$

Approximate the finitely many sampled continuation effects by induction with error at most η. Use the local operators $M_i = \sqrt{w_i} L_i Q^{-1/2}$ and attach those continuation trees. The Gram sum of (M_i) is exactly I. If $R = Q^{-1/2}$ and $u_\delta = (1-\delta)^{-1/2} - 1$, the resulting effect error is at most

$$\eta + \frac{\delta}{1-\delta} + 2u_\delta + u_\delta^2.$$

Indeed the continuation error, before conjugation, lies between $-\eta(Q \otimes I)$ and $\eta(Q \otimes I)$, so after conjugation it is at most η; the remaining terms bound $(R \otimes I) C_j (R \otimes I) - E_j(h)$. Bob's case is identical. Small enough η, δ at each finite induction step give a finite complete tree approximating the finite-horizon report as accurately as desired. Sampling the whole tuple retains outcome-dependent continuation policies.

It remains to approximate an infinite-transcript report. Let $d = ab$. Trace preservation gives consistent probability laws $\mathbb{P}_\rho$ on finite prefixes and hence a unique full law. Its affine dependence on ρ, extended homogeneously to positive inputs, gives event effects by finite-dimensional trace duality. Use the faithful reference I/d with law μ. Since $\rho \le d(I/d)$, positivity gives $\mathbb{P}_\rho(B) \le d\mu(B)$ for every event B. Let $\mathcal{F}_N$ be the first-N-record sigma algebra and set $q_{N,j} = \mathbb{E}_\mu[\mathbf{1}_{J=j} \mid \mathcal{F}_N]$. Choose Borel finite-history versions and report a smallest maximizing index J_N. Upward martingale convergence implies $\mu(J_N \ne J) \to 0$. The effect difference is bounded uniformly by

$$\|F_{N,j} - F_j\|_{\mathrm{op}} \le d\mu(J_N \ne J).$$

Combine this with a finite complete tree approximation of error $1/N$. Appending local orthonormal-basis measurements at all terminal nodes gives literal product-bra leaves without changing the report effects. □

A report completed only at (3) causes no restriction in this lemma's application. Choose one Borel representative under its prepared-reference law. For every density ρ on AB, $V_\chi \rho V_\chi^\dagger \le 4\rho_*$. Positivity of the event operations implies domination of the corresponding full Born laws. The same Borel representative therefore agrees with the completed report for every charged density input. Apply Lemma 2 to that representative on the whole augmented finite input. This supplies effects on the whole input while retaining the required physical probabilities on the resource code.

3 Exact effects and determinant costs

Use local Schmidt coordinates for the charged resource, retaining all its nonzero coefficients:

$$|\chi\rangle = \sum_{k=1}^{r} \sqrt{\lambda_k}|k,k\rangle, \qquad \lambda_1 \geq \cdots \geq \lambda_r > 0, \qquad \sum_k \lambda_k = 1.$$

Set $x = \lambda_1$ and $D = \mathrm{diag}(\sqrt{\lambda_k})$. Passing to Schmidt coordinates is a local isometry on the support and preserves the entire reduced entropy. Rectangular original coefficient matrices and zero Schmidt coefficients are included: their local support embeddings, or their zero padding into a common square space, preserve normalization and all nonzero reduced eigenvalues.

Write $E_j \in M_2(\mathbb{C})$ for the coefficient matrix of e_j. The tetrahedral identities give

$$\langle e_i|e_j\rangle = \frac{1 + 3\mathbf{n}_i \cdot \mathbf{n}_j}{4} = \delta_{ij}.$$

The matrix of s has both singular values $1/\sqrt{2}$; multiplying it by $(I + \sqrt{3}\mathbf{n}_j \cdot \boldsymbol{\sigma})/2$ gives squared singular values $(2 \pm \sqrt{3})/4$. Consequently

$$|\det E_j| = \frac{1}{4}, \quad |e_j\rangle = \tfrac{1}{2}|s\rangle + \tfrac{\sqrt{3}}{2}|t_j\rangle, \quad |t_j\rangle = (\mathbf{n}_j \cdot \boldsymbol{\sigma} \otimes I)|s\rangle, \quad |\langle t_j|e_j\rangle|^2 = \frac{3}{4}. \tag{7}$$

Each t_j is normalized and maximally entangled because $\mathbf{n}_j \cdot \boldsymbol{\sigma}$ is unitary.

Suppose effects F_j on $AA' : BB'$ satisfy

$$F_j \geq 0, \qquad \sum_j F_j = I, \qquad V_\chi^\dagger F_j V_\chi = P_j. \tag{8}$$

For $u_j = V_\chi e_j$, positivity of $I - F_j$ gives $F_j u_j = u_j$. The line spanned by u_j reduces the Hermitian F_j, so $F_j \geq |u_j\rangle\langle u_j|$.

If the F_j are separable, compressing their resource factors onto $|1,1\rangle$ yields separable two-qubit effects $G_j \geq xP_j$ with $\sum_j G_j = I$. For a positive separable two-qubit operator G and a maximally entangled unit vector t, $2\langle t|G|t\rangle \leq \mathrm{Tr}\, G$: it holds for a product projector by its Schmidt overlap and hence for every positive sum. Equation (7) therefore gives

$$4 = \sum_j \mathrm{Tr}\, G_j \geq \sum_j 2x|\langle t_j|e_j\rangle|^2 = 6x, \qquad x \leq \frac{2}{3}. \tag{9}$$

For any finite product decomposition $F_j = \sum_v |a_{jv}, b_{jv}\rangle\langle a_{jv}, b_{jv}|$, absorb positive weights into the vectors. Let $A_{jv}, B_{jv} \in M_{2,r}(\mathbb{C})$ be their coefficient matrices with columns $a_{jv,k}, b_{jv,k}$. Compressing a positive sum to the rank-one P_j forces

$$A_{jv} D B_{jv}^T = c_{jv} E_j, \qquad c_{jv} = \langle u_j|a_{jv}, b_{jv}\rangle, \qquad \sum_v |c_{jv}|^2 = 1. \tag{10}$$

The identity $F_j u_j = u_j$ also gives

$$\sum_v \overline{c_{jv}} a_{jv,1} b_{jv,1}^T = \sqrt{x} E_j. \tag{11}$$

Define

$$T_{jv} = \sum_{k \geq 2} \sqrt{\lambda_k} a_{jv,k} b_{jv,k}^T = c_{jv} E_j - \sqrt{x} a_{jv,1} b_{jv,1}^T, \quad L_{jv} = \sqrt{x} b_{jv,1}^T E_j^{-1} a_{jv,1}.$$

Taking the trace of E_j^{-1} times (11) yields $\sum_v \overline{c_{jv}} L_{jv} = 2x$. The two-dimensional rank-one determinant formula implies

$$\det T_{jv} = (\det E_j) c_{jv}(c_{jv} - L_{jv}).$$

If $c_{jv} = 0$, the tail matrix has rank at most one and determinant zero. Otherwise multiply by $\overline{c_{jv}}/c_{jv}$, sum over v, and use (10). The triangle inequality gives

$$\sum_v |\det T_{jv}| \geq \frac{1}{4}|1 - 2x|, \qquad \sum_{j,v} |\det T_{jv}| \geq |1 - 2x|. \tag{12}$$

This algebraic conclusion holds for every product decomposition of exact effects.

4 The LOCC bound and passage to limits

For a finite complete tree, refine all terminal local outputs to product bras. Put $t = 1 - x$ and suppose $t > 0$. Run that same complete tree on

$$|\tau\rangle = \frac{1}{\sqrt{t}} \sum_{k \geq 2} \sqrt{\lambda_k} |k, k\rangle, \qquad |\Phi_2\rangle_{AR_A} \otimes |\Phi_2\rangle_{BR_B},$$

where R_A, R_B are untouched references local to their respective parties. The only initial interparty entanglement is in τ. The reference coefficient matrix at leaf (j, v) is $\overline{T_{jv}}/(2\sqrt{t})$. Its probability is $\|T_{jv}\|_F^2/(4t)$ and its normalized two-qubit concurrence is $2|\det T_{jv}|/\|T_{jv}\|_F^2$. Zero-probability leaves contribute zero. Thus its probability-weighted concurrence is $|\det T_{jv}|/(2t)$.

For a pure bipartite state define the I-concurrence [4]

$$C_I(\psi) = \sqrt{2(1 - \operatorname{Tr} \psi_A^2)}.$$

This is an average LOCC monotone. The map $\rho \mapsto 1 - \operatorname{Tr} \rho^2$ is concave, and composing with the increasing concave square root preserves concavity. At each local measurement, the opposite party's average reduced state is unchanged, so Jensen's inequality bounds the average new concurrence by the previous one. Its two reduced states have the same nonzero spectrum, so either party may act. Local pure ancillas leave it unchanged. This is the pure-state concavity criterion [5], proved here for the quantity used. On two-qubit pure states it equals the determinant concurrence above.

With $\mu_k = \lambda_k/t$ for $k \geq 2$, monotonicity gives the bound

$$\sum_{j,v} |\det T_{jv}| \leq 2t \sqrt{2\left(1 - \sum_{k \geq 2} \mu_k^2\right)} = 4 \sqrt{\sum_{2 \leq k < \ell} \lambda_k \lambda_\ell}. \tag{13}$$

The tree need not implement the EJM on τ; only its completeness is used in this reference experiment.

We now preserve (13) when effects become exact only in a limit. On unit product vectors $\alpha = a \otimes b$ in the fixed augmented input, form the compact set of augmented moments

$$\mathcal{K} = \{(|\alpha\rangle\langle\alpha|, |\det T(\alpha)|) : \|a\| = \|b\| = 1\},$$

where $T(\alpha)$ contracts the resource tail. If a product projector has weight w, its effect and determinant cost are both multiplied by w. The weights in a decomposition of outcome j sum to $\operatorname{Tr} F_j$, which is bounded by the augmented input dimension. Its augmented moment lies in $\operatorname{Tr} F_j \operatorname{conv} \mathcal{K}$. This

set of possible effect/cost pairs is compact, including trace zero. Along a common subsequence for the four outcomes, convergent effects therefore have convergent determinant costs. Carathéodory's theorem represents each limiting moment by a finite product decomposition with the same limiting cost. The total cost still satisfies (13). If the limiting compressed effects are P_j, apply (12) to these particular decompositions. Separability of the effects is also preserved by this compactness. Hence both bounds hold together and, when $x > 1/2$,

$$(2x-1)^2 \le 16 \sum_{2\le k<\ell} \lambda_k \lambda_\ell. \tag{14}$$

For an exact physical protocol, Lemma 2 supplies the finite trees needed in this argument. For a sequence in (4), apply that lemma to each Borel report representative, choosing a tree with effect error at most $1/k$. Every effect is bounded between 0 and I on the same finite augmented input, so take a convergent subsequence. Its compressed effects equal P_j: convergence on every system density state in (4) identifies the compressed Hermitian matrices by trace duality. Apply the same augmented-moment argument. This proves (9) and (14) in the exact and operational cases. In particular rank one is excluded by (9), justifying $t > 0$ whenever the tail reference test is needed.

5 From the spectrum bounds to one bit

We complete the entropy estimate explicitly. Write h_2 for binary entropy. If $x \le 1/2$, then $H(\lambda) \ge -\log_2 x \ge 1$. Otherwise (9) gives $1/2 < x \le 2/3$. Put

$$z = 2x-1, \qquad t = 1-x, \qquad c = \frac{z}{2t} \in (0,1/2], \qquad F(c) = h_2\left(1-\sqrt{1-c^2/2}\right).$$

Equation (14) bounds the normalized tail purity by $P = \sum_{k\ge 2} \mu_k^2 \le 1 - c^2/2$. If its largest entry m exceeds $1/2$, then $m^2 \le P$ and grouping that entry against the rest gives $H(\mu) \ge h_2(m) \ge h_2(\sqrt{1-c^2/2})$. If $m \le 1/2$, then $H(\mu) \ge 1$, which implies the same bound. Thus

$$H(\mu) \ge F(c). \tag{15}$$

The following elementary inequalities suffice:

$$F(c) \ge \tfrac{3}{2}c^2 \qquad (0 \le c \le 1/3), \tag{16}$$

$$F(c) \ge \tfrac{4}{3}c^2 \qquad (0 \le c \le 1/2), \tag{17}$$

$$1 - h_2(x) \le \tfrac{3}{4}(2x-1)^2 \qquad (1/2 \le x \le 2/3). \tag{18}$$

For $c > 0$ set $q = 1 - \sqrt{1-c^2/2}$. Then $c^2 = 4q(1-q/2)$ and $-\ln(1-q) \ge q$ gives

$$h_2(q) \ge \frac{q[\ln(1/q) + 1 - q]}{\ln 2}.$$

For $c \le 1/3$, one has $q < 1/32$, so $\log_2(1/q) \ge 5$ and $(1-q)/\ln 2 \ge 1$; hence $h_2(q) \ge 6q \ge (3/2)c^2$. For $c \le 1/2$, one has $q < 1/15$. The ratio

$$R(q) = \frac{\ln(1/q) + 1 - q}{4 \ln 2 (1 - q/2)}$$

decreases on $(0,1)$: its derivative has the sign of $-1/q + \frac{1}{2}\ln(1/q) < 0$. Consequently

$$\frac{F(c)}{c^2} \ge R(1/15) = \frac{\log_2 15 + (14/15)/\ln 2}{58/15} > \frac{31/8 + 4/3}{58/15} = \frac{625}{464} > \frac{4}{3}.$$

Here $\log_2 15 > 31/8$ follows from $15^8 > 2^{31}$, and $\ln 2 < 7/10$ implies $(14/15)/\ln 2 > 4/3$. The inequalities at $c = 0$ follow directly.

For $0 \le z \le 1/3$, expansion of the logarithms yields

$$1 - h_2((1+z)/2) = \frac{1}{\ln 2} \sum_{n \ge 1} \frac{z^{2n}}{2n(2n-1)} \le \frac{z^2}{\ln 2} \left(\frac{1}{2} + \frac{z^2}{12(1-z^2)} \right) \le \frac{49 z^2}{96 \ln 2} \le \frac{3}{4} z^2.$$

For these constants, the first two terms and a geometric tail bound in $\ln 2 = 2 \sum_{n \ge 0} [(2n+1) 3^{2n+1}]^{-1}$ give $56/81 < \ln 2 < 56/81 + 1/540 < 7/10$ and $56/81 > 49/72$. This proves (16)–(18) without numerical estimates.

Finally $H(\lambda) = h_2(x) + tH(\mu)$. If $x \le 5/8$, then $c \le 1/3$ and $t \le 1/2$, so (15) and (16) give

$$tH(\mu) \ge \frac{3z^2}{8t} \ge \frac{3}{4} z^2.$$

If $5/8 \le x \le 2/3$, then $t \le 3/8$, and (17) gives

$$tH(\mu) \ge \frac{z^2}{3t} \ge \frac{8}{9} z^2 \ge \frac{3}{4} z^2.$$

Combining either bound with (18) proves $H(\lambda) \ge 1$.

6 Attainment

Supply $|\Phi_2\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, whose reduced entropy is one bit. Alice measures her input qubit and her resource half in a Bell basis and sends the two-bit outcome to Bob. Bob's corresponding Pauli correction transfers Alice's input state, including all its correlations with his input, to his resource half. He measures the two qubits locally in the orthonormal basis (e_j) and sends j to Alice. For every density input the resulting shared label probabilities are $\mathrm{Tr}(P_j \rho)$, so the protocol implements (2) exactly. It is a finite complete physical LOCC protocol. A constant sequence gives an operational implementation with the same resource. Both cost sets therefore contain 1, and the established lower bound on every member proves Theorem 1.

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